

Water level Indicator

PROJECT REPORT

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BECE102L – Digital Systems Design

By

M.Hariharan Reg. No-22BEC1126
Udit Krishna Reg. No-22BEC1080
KL Krishna Chaitanya Reddy Reg. No-22BEC1215

Slot: B1+TB1

Name of faculty: Dr. K. CHITRA

SCHOOL OF ELECTRONICS ENGINEERING



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I extend my thanks to all my well-wishers and all those who have contributed directly and indirectly for the completion of this work

ABSTRACT

Water level indicators play a crucial role in various applications, such as water tanks, reservoirs, and industrial processes, to ensure efficient water management and prevent overflows or shortages. This project proposes a simple and cost-effective water level indicator system based on NOT gates, a fundamental component of digital logic circuits. The Water Level Indicator Using NOT Gates project aims to provide a reliable and easily deploy-able solution for monitoring water levels in a container. When water levels rise or fall, the NOT gates, which, in turn, control LED indicators or alarms to signal the water level status. The simplicity and effectiveness of this design make it an excellent choice for applications where water level monitoring is essential

In conclusion, the Water Level Indicator Using NOT Gates project provides a practical and efficient solution for water level monitoring, ensuring that water resources are managed effectively while minimizing the risk of overflow or depletion. This project demonstrates the application of fundamental digital logic components in a real-world scenario.

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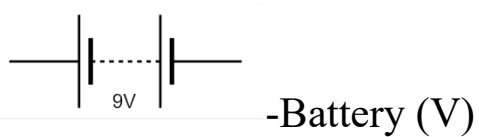
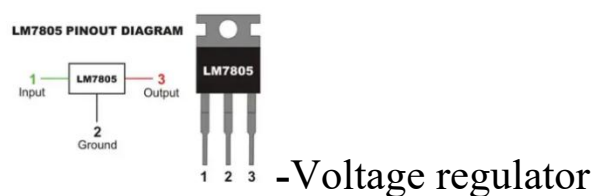
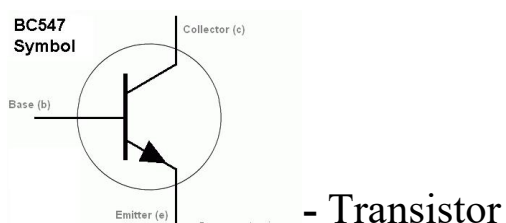
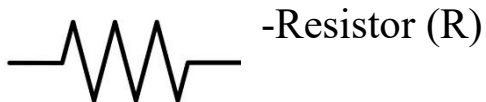
Table 1:

NOT GATE

INPUT (A)	OUTPUT (A')
1	0
0	1

LIST OF SYMBOLS AND ABBREVIATIONS

SYMBOLS



CHAPTER – 1

INTRODUCTION

1.1 MOTIVATION

The goal of a water level indication project is to solve several real-world requirements and concerns pertaining to safety and water management. The following are the main reasons to carry out a project of this kind:

Water Conservation: Efficient use of water resources is critical in today's world, where water scarcity is becoming an increasing concern. Monitoring water levels in reservoirs, tanks, or wells helps in preventing wastage and encourages responsible water use.

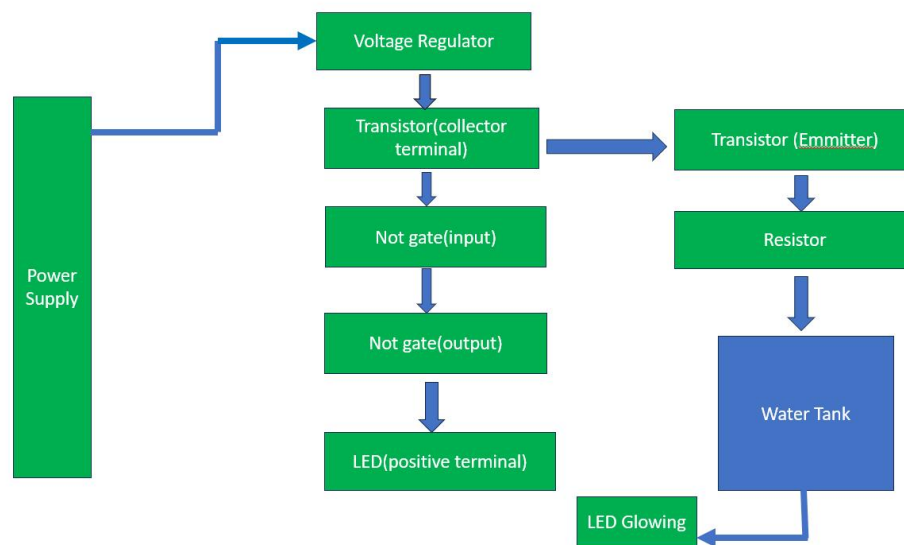
Preventing Overflows: Overflows from water storage containers can lead to property damage, flooding, and even environmental hazards. A water level indicator system can provide an early warning to prevent such incidents.

Cost Savings: By monitoring and maintaining water levels, users can optimize water consumption and reduce water bills. This can be particularly relevant for homeowners, farmers, and industries that rely on water for their operations.

1.2 Innovative Content in the Project

As we all know water is a precious that is essential for life. However water scarcity is a growing problem around the world .One way to conserve the water is to use it efficiently.So by using this water level indicator we can efficiently use our water and prevent wastage.Traditional water level indicators are often mechanical devices that use sensors to measure the water level in water tank.These devices are often bulky and difficult to install.They can also be inaccurate,especially if they are not properly calibrated .To overcome this issue we have designed this project which involves simple connections using NOT gate,Transistors,Voltage regulators,Resistors and few wires and the connections are simple .The benefits of this project include Increased accuracy,Easier installation,Lower cost,Greater versatility as compared to traditional water level indicator.

1.2 Block Schematic



1.2.1. Module descriptions

LED(different colors)

Not Gate: NOT gates are digital logic gates with a single input and an inverted output. In this module, NOT gates are used to convert the signals received from the water level

BC547 transistor: Transistors are used as switching devices in this module. Each transistor is connected to the output of a NOT gate. When the NOT gate outputs a logical high (indicating water presence), the transistor is turned on.

Voltage regulator: A voltage regulator is a component of the power supply unit that ensures a steady constant voltage supply through all operational conditions. It regulates voltage during power fluctuations and variations in loads

Resistor: It is an electrical component that limits or regulates the flow of electrical current in an electronic circuit

Battery: A battery is a device that stores energy and then discharges it by converting chemical energy into electrical energy

Connecting wires (jumper wires) - (20-30)

Breadboard

Buzzer

1.2.2. Cost of the Project

The total cost of a water level indicator project is 400 approximately.

COMPONENTS	COST
Jumper wires (35)	105
Voltage regulator	60
Resistors	20
Not Gate(7404)	19
LED's	10
Battery	28
Transistor	40
Breadboard	95

1.2.3. Advantages and Disadvantages

Advantages:

Cost-Effective: This type of water level indicator is generally affordable, making it accessible for various applications, including home use and small-scale projects.

Simplicity: The use of basic electronic components such as NOT gates and transistors makes the project relatively simple.

Low Power Consumption: When designed efficiently, this type of water level indicator can consume minimal power, making it suitable for battery-powered applications.

Customization: You can customize the project to fit your specific needs by adjusting the number of transistors and output indicators.

Disadvantages:

Limited Precision: This type of water level indicator typically offers limited precision, with discrete "water present" or "water absent" indications. It may not provide fine-grained information about the water level.

Maintenance: The project may require periodic maintenance to ensure that electronic components continue to function correctly.

No Remote Monitoring: Without additional components or micro-controllers, the project may lack remote monitoring or data logging capabilities, which can be a disadvantage in certain applications.

1.4 Overview of the Report

The water level indicator project is a simple and useful project that can be implemented using a NOT gate and a transistor. The project aims to detect the water level in a tank and indicate it using LEDs. The circuit works by using a transistor as a switch to turn on the LED when the water level reaches a certain point. The NOT gate is used to invert the signal from the transistor, which is then used to turn on the LED. This report provides a detailed description of the project, including the working principle and full description of the circuit. It also includes a list of materials required for the project. This report provides a detailed presentation of the water level indicator project using a transistor and other electronic components.

CHAPTER-2

LITERATURE SURVEY

1) Pandey et.al, 2011[1]-Explained Agricultural land management practices are compatible with the preservation of water resources. Hydrological diagnoses are needed in order to choose the alternative land uses, cultivation practices and their spatial arrangements.

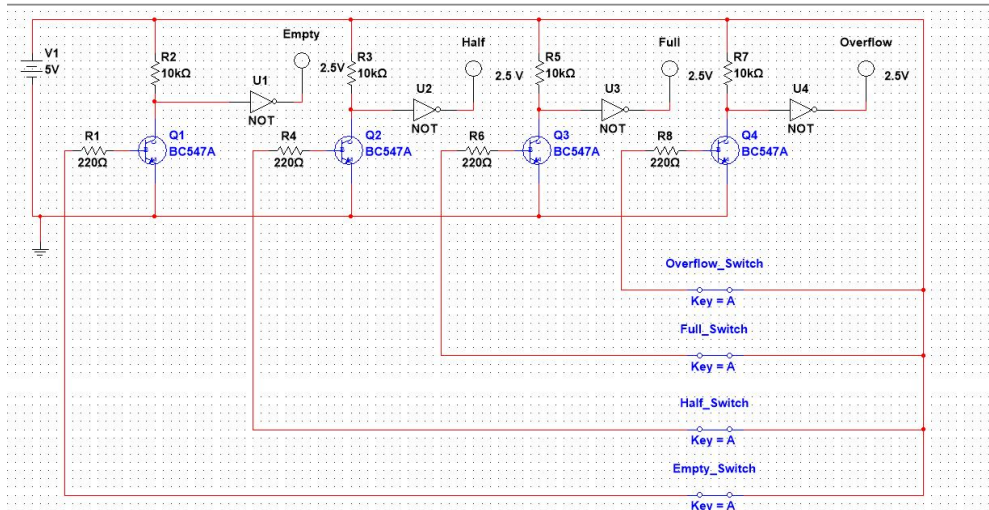
2) Jaehyoung Yu Harnsoo Han, 2006[2] explained the monitoring water level in a river or in a reservoir is important in the applications related to agriculture, flood prevention and fishing industry etc. The schemes developed for measuring water level can be categorized as 4 types based on measuring features: pressure, supersonic waves, heat, image.

3) M. H. Khairuddin, N. A. A. Aziz, A. A. bin Azmi, and M. A. bin Mohd Yusof, "Design and Implementation of a Water Level Monitoring System Using Float Switch and Relay," International Journal of Applied Engineering Research, vol. 11, no. 22, pp. 10169–10173, 2016.

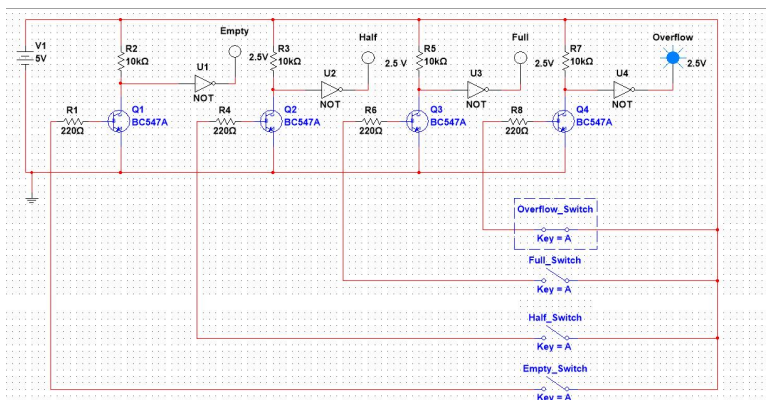
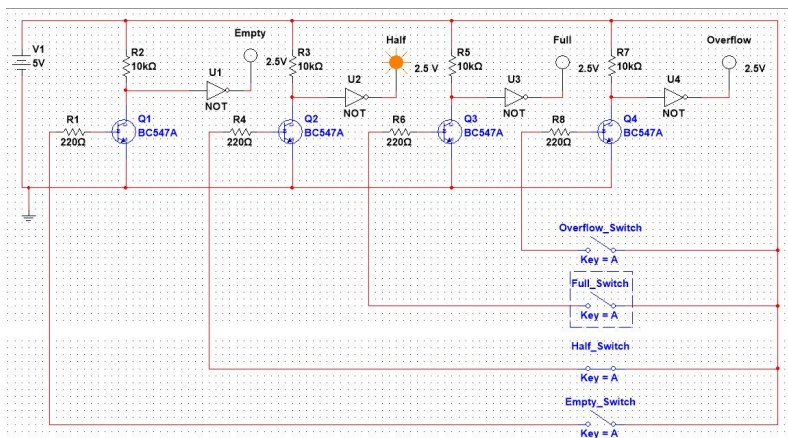
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DESIGN AND SIMULATION

3.1 Design of the Project



3.2 Screenshots of simulated results



CHAPTER -3

HARDWARE

4.1 Hardware Description

The hardware components used in the "Water Level Indicator" project are essential for detecting water levels and conveying this information to the user through visual and audible indicators.

4.1.1 Working

The components used are IC 7404, BC 547 TRANSISTOR, RESISTOR – 10K OHM, 220 OHM, BREADBOARD, WIRES, LED, 11.1V BATTERY, 7805 MOSFET, NOT GATE, BUZZER

In this project “water level indicator” there are 3 main conditions:

Condition 1:

Water is not available in the tank:

By looking at the circuit we can get to know about the connections. Coming to the first condition where the tank is empty when the base transistor connection wire and the 5V connection wire is dipped in the water it gets short circuited and transistor gets open circuited and there is voltage gets dropped in the 10k ohm resistor and at the collector point there is 0 voltage which acts as a input for the NOT gate and gives output 1 where the LED light glows yellow which indicates that the tank is empty.

CONDITION 2:

WATER IS HALF OR FULL IN THE TANK:

Coming to the second condition where the water is half or full in the tank. Lets first see what happens when there is half water in the tank. When the base

transistor connection wire is dipped in water along with base transistor connection and 5V connection wire because only after these connections are dipped, water can touch transistor base connection wire. So when all the three wires are dipped in water they get short circuited 2nd transistor becomes open and there is voltage drop across 10k ohm and there is 0 voltage at collector connection wire which is the input for the NOT gate which gives 1 as output and the LED glows in the colour of orange which indicates that the water in the tank is half. This is the same process when the water in the tank is full where the LED light glows in green colour.

CONDITION 3: WATER IS OVERFLOWING IN THE TANK:

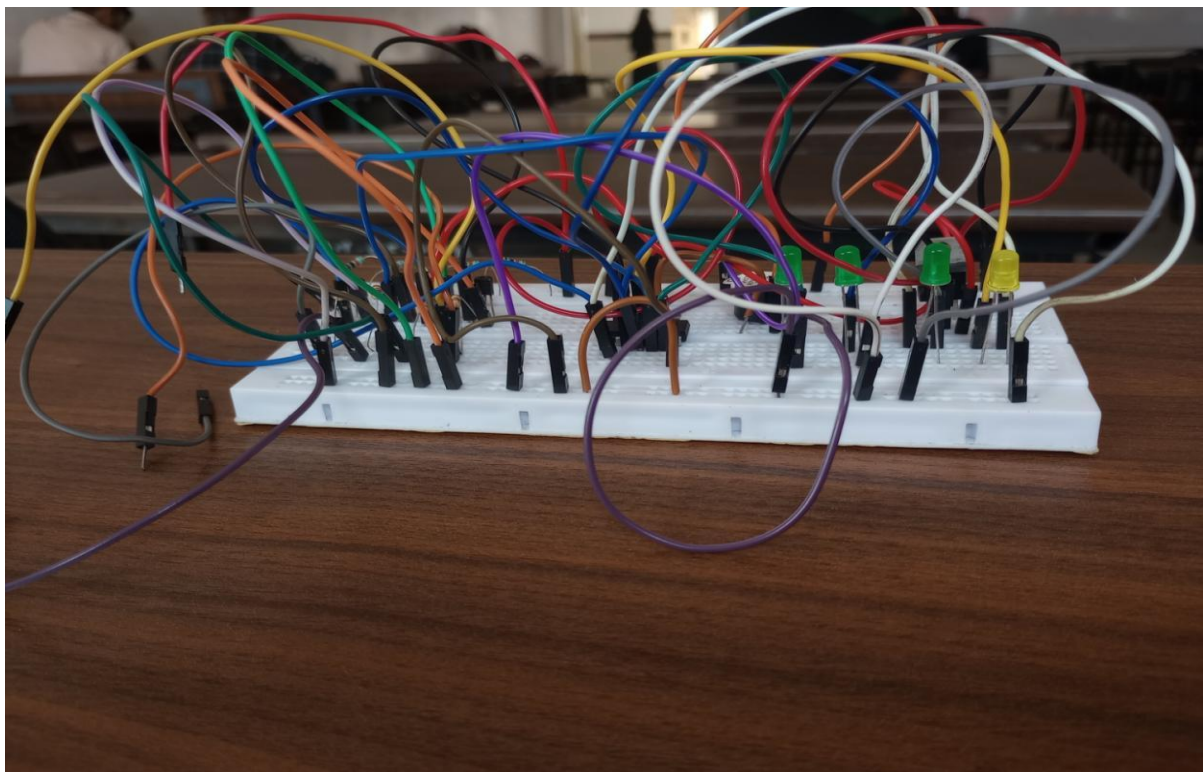
Coming to the third and final condition where the water in the tank is overflowing. So when there is an overflow in the tank it means that all the all base transistor wires (Q1,Q2,Q3) and 5V are dipped in the water as it an overflow condition and even base transistor wire is also dipped in the water and which makes the transistor 4 open and there is voltage drop across 10k ohm resistor and the collector wire voltage is zero which is the input for the NOT gate and gives output 1 where the LED light glows in blue colour also the buzzer gives the sound which indicates that there is overflow of water in the tank.

4.2 Results and Discussion

Coming to the results and discussion, we have successfully the device (water level indicator) that can accurately measure the water level in bottle. The device is easy to install and operate and it is relatively inexpensive and the project is very simple and highly efficient in conserving water. The device was

able to accurately measure the water level in the bottle. This project also has various applications such as Domestic water conservation, Industrial water management, Agricultural irrigation, Environmental monitoring. As the technology continues to develop, water level indicators are likely to become affordable and widely used. So this project is really helpful in conserving water. It should be marketed to wider audience, should be made more affordable also this project should be developed to make it more accurate and versatile. It should be installed in all homes for knowing the amount of water that has been consumed and minimize the usage of water just for the needs and not to use excess of water and the government should take initiatives to make this device widely available to people. I hope this project has the potential to make a significant impact on water level conservation efforts around the world.

4.4 Screenshots of hardware used and results



CONCLUSION AND FUTURE SCOPE

The water level indicator is the best electronic starter device indicating the water level and saves water correspondingly. The led blinks according to their level of water reaches to them and indicates the buzzing sound to stop pouring when it reaches its final limit.

The system proved to be reliable in detecting water levels, providing accurate and timely indications of water presence or absence.

The low power consumption of the system aligns with its goal of being an energy-efficient solution.

The "Water Level Indicator" project opens up several avenues for future enhancements and modifications:

Consider improving the system's accuracy by incorporating additional sensors and logic components to provide more detailed information about water levels, enabling users to monitor intermediate levels. Explore the integration of advanced features, such as remote monitoring and control. Incorporating micro-controllers and wireless communication technologies would extend the system's capabilities.

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